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How to construct 'normal ranges' for fetal variables

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ABSTRACT

A method to estimate reference intervals which depend on gestational age is described. The approach is based on standard statistical techniques, so implementation should be straightforward with most statistical packages. The reference limits are defined by simple formulae which may be plotted as smooth curves. The method is illustrated with analyses of three real datasets arising from different aspects of fetal monitoring.

INTRODUCTION

Despite the efforts made by authors such as Altman and Chitty[1] and Royston[2], incorrect methods are still being used to estimate age-specific reference intervals ('normal ranges') for fetal measurements of all kinds. The choice of suitable methodology is crucial, because inaccurate centiles obtained from an inferior method may mislead the obstetrician as to the true state of health or development of the fetus and increase the chance of suboptimal clinical care. We prefer the term 'reference interval' rather than 'normal range' since the latter implies that any observation outside the range indicates an 'abnormality', which is not necessarily the case.

A reference interval is the range of values encompassed by a pair of symmetrically placed extreme centiles, such as the 2.5th and 97.5th centiles for a 95% interval. Values which lie outside the reference limits are regarded as unusual or extreme and may indicate the presence of a disorder such as intrauterine growth restriction (IUGR). Indeed, suspected IUGR is often *defined* in terms of fetal size lying below a centile curve such as the 10th or 5th. To construct reference intervals, data are collected from a reference population which usually consists of scans from a representative sample of fetuses from unselected pregnancies. Typically, the status of the pregnancy is determined retrospectively, and there is scope for controversy in deciding which pregnancies should be defined as abnormal and excluded from the reference sample. See Altman and Chitty[3] for a detailed discussion of the design issues in such studies. Here we mention only a few salient points.

It is important to distinguish between cross-sectional and longitudinal studies. In the former, each fetus contributes just one value to the reference sample, whereas, in the latter, each fetus is measured several times. When the measurement is of an aspect of fetal 'biometry' such as abdominal circumference, reference intervals may allow one to assess, at a given gestational age, the size or the growth of the fetus in relation to the reference population. However, cross-sectional data give information only on size, whereas longitudinal data may be used to produce reference intervals either for size or for growth. Correct analysis of longitudinal fetal data is somewhat involved. It is described in statistical terms by Royston[4] and more generally by Royston and Altman[5]. We shall not consider it further here.

When collecting the data, it is essential to record the gestational age precisely to the day as best one can, rather than truncating it to the number of 'completed weeks'. (The well-known problems of pregnancy dating to establish gestational age may also affect the precision of the recorded value, but this is a clinical issue on which we have little comment.) Data which have been truncated to completed weeks may include, at any given week, fetuses which are up to 6 days older than the recorded value. Since many measurements increase rapidly with age, there is potential for considerable error in centile estimates if reference values reported for completed weeks are mistakenly assumed to be for exact weeks. (Not all authors state which version they are using.) If exact ages are unobtainable, ages in completed weeks should be adjusted to avoid bias by the addition of 0.5 weeks. To avoid confusion, we strongly recommend that authors analyze and publish their results for only exact weeks.

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Almost all fetal measurements change with gestational age. In most reported analyses, the mean is modelled as age-varying, but often dependency of the standard deviation of the values on gestational age is not recognized and not allowed for. When it changes at all, the standard deviation almost invariably increases with gestational age, so the result of such neglect is reference intervals which are too wide in early pregnancy and too narrow in the later stages. Furthermore, we believe that in reality the mean and standard deviation will change smoothly with age, so chosen methods of analysis should reflect this requirement. It is still common to see reports of centile estimates which vary in a jagged fashion with age, usually because the authors have calculated means and standard deviations by week of gestation, with no attempt at smoothing.

In this paper, we describe a method which in our experience is sufficiently general to cope with a wide range of fetal measurements available from ultrasound scanning and from investigations such as blood sampling for gases and other analytes. Although the methodology is closely related to that of Altman and Chitty[1], we feel that a paper which gives more detail of the practical implementation may be helpful. The aim is to enable anyone with a basic knowledge of statistical techniques to apply the method to their own data. The approach is based upon regression modelling of both the mean and the standard deviation across gestational age. Some variables are much more easily modelled when transformed to a logarithmic scale, either because of positive skewness in the distribution of the original values (a tendency towards a long 'tail' of high values) or because the standard deviation increases very rapidly with gestational age. Measurements that exhibit this behavior include estimated fetal weight and fetal kidney volume.

The next section describes the steps required to estimate a reference interval, and illustrates specific details using data on fetal head circumference. The 'Examples' section describes the application of the method to a further two measurements. The section on 'Pitfalls' describes problems which arise when modelling is performed by age group and when an inadequate sample size is used. The discussion summarizes the method and suggests how further improvements may be made for dealing with variables that do not have a Gaussian (also known as Normal) distribution, an implicit assumption made by many authors.

METHODS

Details of the method will be illustrated using data from a designed prospective study of fetal size[6]. A large number of ultrasound measurements were made for 610 normal fetuses aged between 12 and 42 weeks. Figure 1 is a scatter plot of head circumference against gestational age in exact weeks for 594 fetuses. From this graph we can see that the head circumference increases in size with increasing gestational age. Though less obvious, it is also useful to note that the standard deviation of head circumference also increases with increasing gestational age.

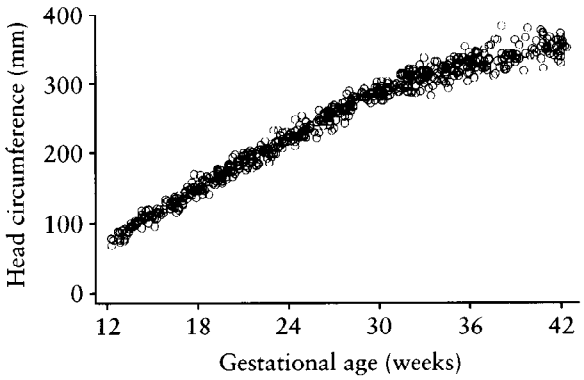


Figure 1 Scatter plot of head circumference vs. gestational age

Statistical model

We assume that, at each gestational age, the measurement of interest has a Gaussian distribution with a mean and standard deviation (SD) and that, in general, both vary smoothly with gestational age. A centile curve is calculated using the well-known formula

$$\text{centile} = \text{mean} + K \times \text{SD} \qquad (1)$$

where K is the corresponding centile of the standard Gaussian distribution. For example, determination of the 10th and 90th centile curves requires that $K = \pm 1.28$. The aim is to find functions that adequately represent how the mean and SD change with age, for then any desired centile curve may readily be calculated by appropriate choice of K in (1).

If the SD increases strongly with age, the logarithm of the measurements will be found to be easier to model than the original values. It is immaterial whether logarithms to base e or base 10 are used. A scatter plot of the logarithmically transformed measurements against gestational age will help to ascertain whether transformation is advantageous. If so, the standard deviation of the transformed values will change far less with age than before. There is a danger that transformation may have induced appreciable negative skewness, indicated by a straggly 'tail' of low values. If so, it is usually better to work with the original values. Centile curves on the original scale are obtained by applying antilogs to the calculated curves. Figure 2 illustrates the data of Figure 1 with head circumference plotted on the natural log scale. One subjective impression may be that the spread of values across age is slightly more constant on the log scale than on the original scale. However, the effect is small, so the analysis will be performed on the original scale.

Regression analysis

Polynomials will often provide adequate curve shapes with which to represent the relationships between fetal variables and gestational age. Least-squares regression analysis may be used to estimate both the mean and SD curves as polynomial functions of gestational age. The following two subsections will clarify the details, but it is worth pointing

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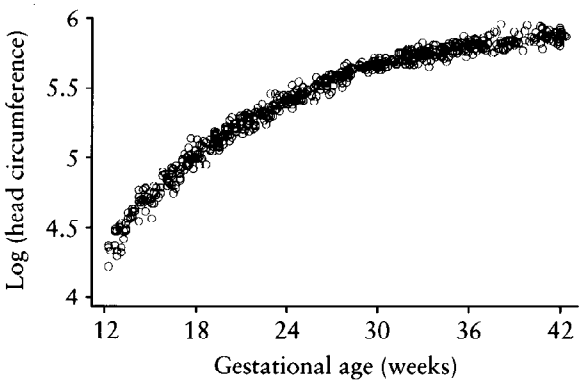


Figure 2 Scatter plot of log transformed head circumference vs. gestational age

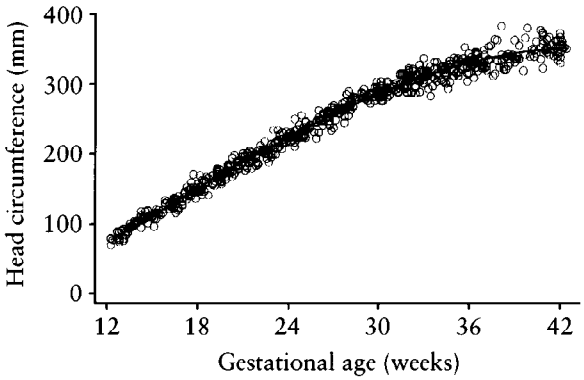


Figure 3 Head circumference vs. gestational age with cubic polynomial fit

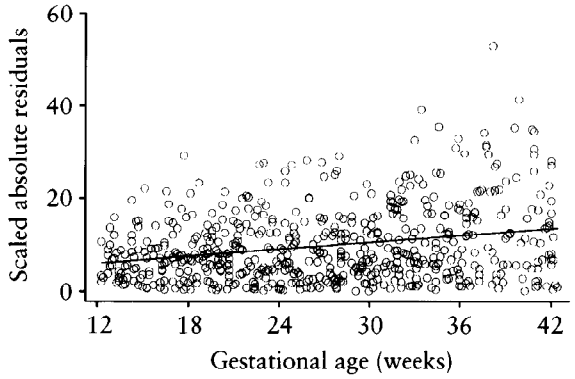


Figure 4 Scaled absolute residuals vs. gestational age with linear fit

out here that the SD curve will almost always require a less complex shape than the mean curve. In most cases, a straight line ($a + b \cdot$ age) will be adequate for the SD.

An approach to choosing the degree of polynomial is as follows. First, a cubic polynomial ($a + b \cdot \text{age} + c \cdot \text{age}^2 + d \cdot \text{age}^3$) is fitted. If the coefficient d is not significantly different from zero (for example, d is less than twice its standard error), a quadratic polynomial is fitted and the same assessment is made of the quadratic coefficient (c). The process is repeated until no further removal of terms is possible. We do not recommend fitting polynomials of degree higher than cubic because often the fitted curves will exhibit unrealistic features such as waviness or sharp deviations at extreme ages. However, if the coefficients of the quartic or quintic polynomials are statistically significant, then we recommend fractional polynomials[7]. These are essentially polynomials in which the powers of age need not simply be the numbers 1, 2, 3, . . . but may take a much wider range of values such as −1 (inverse of age) or 0.5 (square root of age). The flexibility of fractional polynomials is such that, even when a cubic curve fit looks reasonable, a fractional polynomial with fewer terms often fits better. See Appendix A for further details, including a procedure for determining the best fitting model.

Curves for the mean

With the use of the procedure described in the previous subsection, the mean is estimated by the fitted values from an appropriate polynomial regression curve of the measurement of interest on gestational age. For the head circumference data, a cubic polynomial is found to fit significantly better than a quadratic. Its equation is as follows:

$$\text{mean} = -70.69 + (10.21 \times \text{age}) + (0.2039 \times \text{age}^2) + (-0.004932 \times \text{age}^3)$$

Investigation of fractional polynomials gives the formula

$$\text{mean} = -29.32 + (1.705 \times \text{age}^2) + (-0.3981 \times \text{age}^2 \times \log \text{age}).$$

The fit of the fractional polynomial curve is similar to that of the cubic curve and we will proceed using the latter. The fitted curve is shown in Figure 3.

Curves for the standard deviation

When the measurement has approximately a Gaussian distribution, the fitted values following regression of the 'scaled absolute residuals' on age are an estimate of the SD curve. The 'scaled absolute residuals' are the residuals (differences between the measurements and the estimated curve for the mean) with the sign removed and multiplied by 1.25. If the scaled absolute residuals appear to show no trend with gestational age, the SD is estimated as the standard deviation of the unscaled residuals (measurement minus the estimated mean curve). If there is a trend, then polynomial regression analysis is needed to estimate an appropriate curve in the same way as for the mean.

Figure 4 shows the scaled absolute residuals for the cubic mean curve of the previous section plotted against gestational age. The scaled absolute residuals have considerable scatter. Nevertheless, the polynomial regression procedure described above suggests that the relationship may be adequately represented by a straight line. The fitted line is superimposed on the data in Figure 4. Its equation is

$$\text{SD} = 2.764 + (0.2565 \times \text{age}).$$

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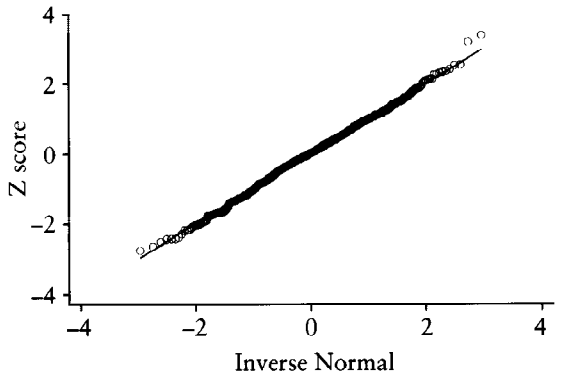


Figure 5 Normal plot of Z scores from fitted model for head circumference data

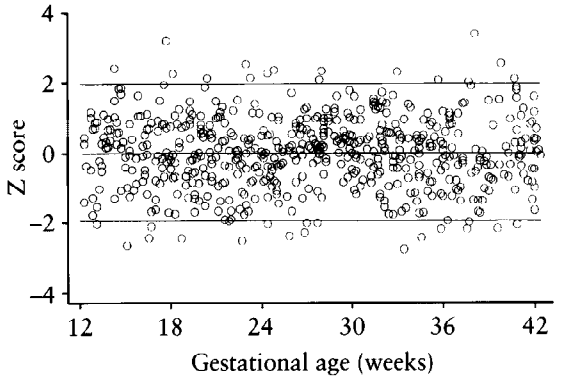


Figure 6 Scatter plot of Z scores vs. gestational age, with tramlines at −1.96, 0, 1.96, from fitted model for head circumference data

Checking the model

The standard deviation scores or Z scores are a useful tool in assessing model fit. The Z scores are defined as

$$Z = \frac{\text{measurement} - \text{mean}}{\text{SD}} \qquad (2)$$

They represent the measurements expressed on a standard Gaussian scale, with the mean and standard deviation both adjusted for gestational age.

A Normal probability plot of the Z scores gives a subjective impression of the degree to which the values follow a Gaussian distribution. The ordered Z scores, when plotted against the corresponding ordered values for a sample from a standard Gaussian distribution with the same sample size, should lie roughly on a straight line. This may be checked formally by the Shapiro–Francia W' test. The W' statistic is the square of the correlation coefficient from the data in the Normal plot. (See Altman's book[8] for further details.) If the p value from the test is statistically significant (say, < 0.05), a more complex curve for the mean or SD may be needed (see Discussion). A plot of the Z scores against age should be randomly scattered about zero. If there is a preponderance of values above or below zero in particular zones of the plot, the mean curve may require modification. Further, approximately 90% of the values should lie between the limits ± 1.645. If there is a deficiency or an excess of values outside these limits for some gestational ages, a higher-order polynomial curve for the SD may be needed.

The Z scores were calculated for the head circumference data based on the cubic mean and linear SD curves. The Normal plot of these values (Figure 5) appears fairly linear and the p value for the Shapiro–Francia test is 0.43, suggesting no departure from a Gaussian distribution. Figure 6 shows the Z scores plotted against gestational age. They appear to be randomly scattered with no obvious patterns, trends or excesses.

The final model

Once a satisfactory model has been determined, the centile curves for the desired reference interval may be calculated by substituting the expressions for the mean and SD into equation (1). Remember to antilog the curves if the measurement was modelled on the log scale. The Z score for any new individual may be calculated using equation (2) and its centile obtained from the inverse standard Gaussian distribution.

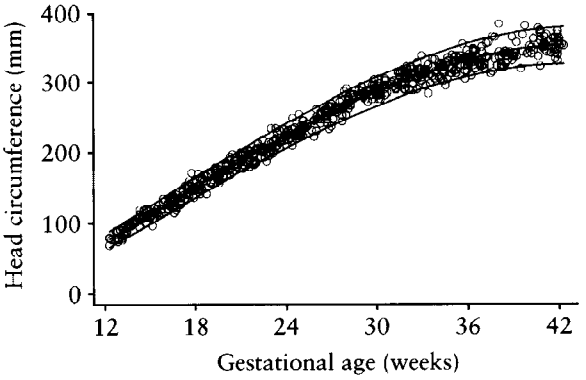


Figure 7 95% reference interval and mean for final model fitted to head circumference data

The model for the head circumference which comprises a cubic polynomial for the mean and a straight line for the SD appears to be satisfactory. (Chitty and colleagues[6] obtained a slightly different model from that described here, but their final curves are almost identical to our curves shown here. In practice, it is commonplace to find different models that produce very similar results.) Figure 7 shows the estimated 95% reference interval superimposed on the raw data. From this model, a fetus with head circumference of 300 mm at 30 weeks' gestation lies on the 89.6th centile ($Z = 1.26$), whereas the same value at 33 weeks lies on the 13.9th centile ($Z = -1.09$). Table 1 contains the values of the mean and 95% reference interval for gestational age at 1-week intervals.

EXAMPLES

The following examples illustrate the methods described above, using datasets that represent different fetal parameters.

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Table 1 Mean and 95% reference interval for head circumference calculated for individual values of gestational age

Gestational age (weeks)	*Head circumference* (mm)		*Gestational age* (weeks)	*Head circumference* (mm)	
	Mean	*95% reference interval*		*Mean*	*95% reference interval*
12	73	(61, 84)	27	257	(238, 276)
13	86	(74, 98)	28	267	(248, 287)
14	99	(86, 111)	29	277	(257, 297)
15	112	(99, 125)	30	287	(266, 307)
16	125	(111, 138)	31	296	(275, 317)
17	138	(124, 152)	32	304	(283, 326)
18	151	(136, 165)	33	312	(290, 334)
19	163	(148, 178)	34	320	(297, 342)
20	176	(160, 191)	35	326	(303, 349)
21	188	(172, 204)	36	333	(309, 356)
22	200	(184, 217)	37	338	(314, 362)
23	212	(195, 229)	38	343	(318, 367)
24	224	(207, 242)	39	347	(322, 372)
25	235	(217, 253)	40	350	(325, 376)
26	246	(228, 265)	41	353	(327, 379)

Hemoglobin

Figure 8 shows measurements of fetal hemoglobin concentration against gestational age in a sample of 459 fetuses[9]. A log transformation of hemoglobin makes little difference to the scatter and overall appearance of the plot, so no transformation is needed. Investigation using polynomial regression reveals that a straight line is satisfactory for the mean. There is little evidence of age dependency in the standard deviation and it is therefore estimated as a constant (the residual standard deviation). A Normal plot of the resulting Z scores looks fairly linear, the Shapiro–Francia test having a p value of 0.2. There are no apparent patterns in the plot of the Z scores against gestational age, so there is no evidence that the model fits poorly. We conclude that the model is reasonable. The mean curve and a 95% reference interval are shown in Figure 9.

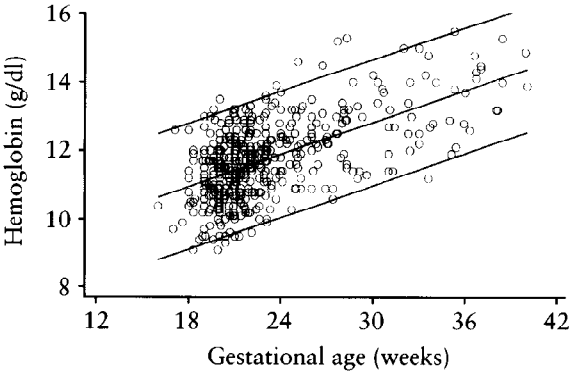


Figure 9 95% reference interval and mean for hemoglobin data

Kidney volume

Figure 10 contains the kidney volume measurements and gestational ages of a sample of 424 fetuses[1]. The effect of applying a log transformation to the kidney volume is illustrated in Figure 11. The transformation dramatically reduces the dependence of the spread of the values on gestational age. The mean may be adequately modelled

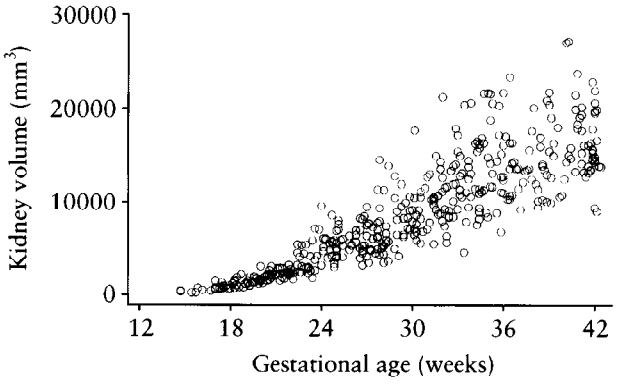


Figure 10 Scatter plot of kidney volume vs. gestational age

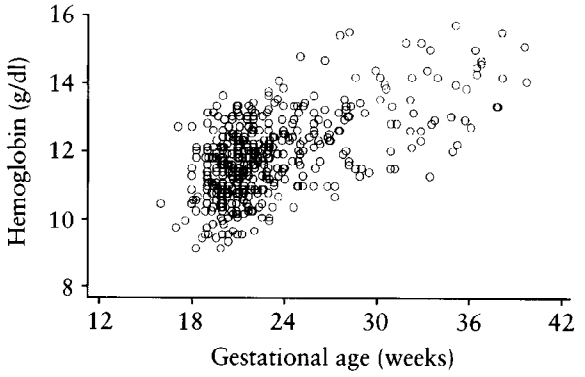


Figure 8 Scatter plot of hemoglobin vs. gestational age

with a cubic polynomial or with a fractional polynomial having the terms log and square-root of age. We somewhat prefer the fractional polynomial on the grounds of simplicity, as it has only two terms as opposed to three for the cubic. The pattern of the scaled absolute residuals suggests that the standard deviation decreases approximately linearly with gestational age. A Normal plot of the resulting Z scores is linear and the Shapiro–Francia test has a p value of 0.6. A plot of the Z scores vs. age (see Figure 12) shows no obvious trends or patterns. The antilogged mean

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and an estimated 95% reference interval are plotted in Figure 13 together with the original observations.

PITFALLS

Grouping by age

Sometimes reference intervals are calculated as the mean plus or minus twice the standard deviation for each week of gestation individually. In Figure 14, these estimates are plotted against the number of completed weeks of gestation for the head circumference data. (To center the gestational age groups, 0.5 weeks has been added to each number of completed weeks.) Clearly the mean does not change smoothly with age, as we would expect on biological grounds. Also the SD (represented by one-quarter of the width of the vertical bands) varies considerably. Further, the added difficulty of truncated gestational ages, as discussed in the Introduction, is present. For example, values of 20 weeks and 20 weeks and 6 days are both classed as 20.5 weeks. These issues are avoided when the mean and SD are each modelled with the use of exact gestational ages and smooth regression functions, as described in the previous sections. Figure 15 compares the curves from the grouped analysis with those from the regression approach.

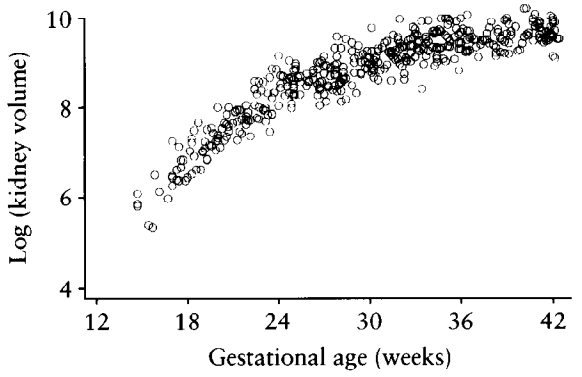


Figure 11 Scatter plot of log transformed kidney volume vs. gestational age

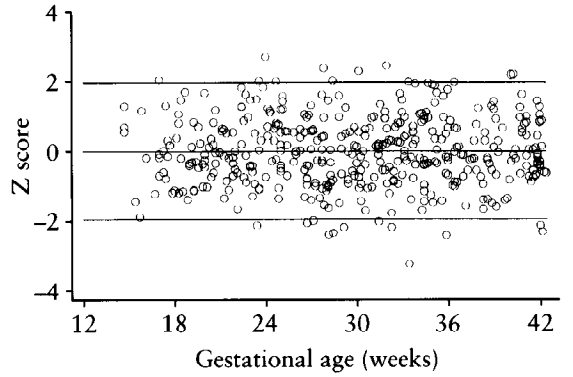


Figure 12 Scatter plot of Z scores vs. gestational age, with tramlines at −1.96, 0, 1.96, from fitted model for kidney volume data

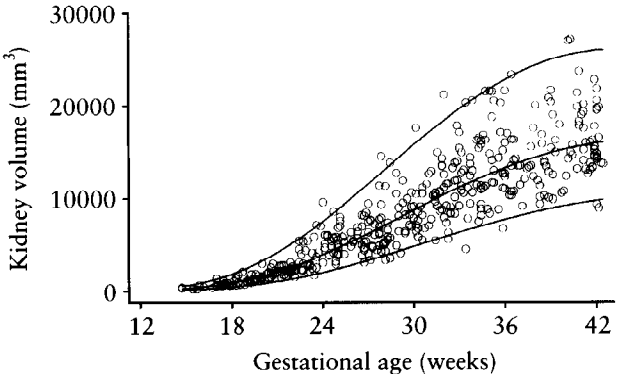


Figure 13 95% reference interval and mean from fitted model for kidney volume data

Inadequate sample size

Imprecise estimates of the reference interval will be obtained when the sample size of the dataset is too small. The precision of the limits of the interval may be estimated from the standard errors of the centile estimates. The standard error (SE) of a given centile curve is given by

$$SE = \sqrt{SE(mean)^2 + K^2 SE(SD)^2}$$

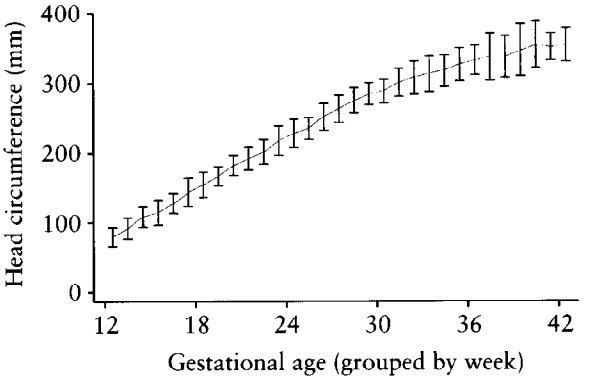


Figure 14 Estimated mean ± 2 standard deviation calculated for data grouped by number of completed weeks of gestation for head circumference data

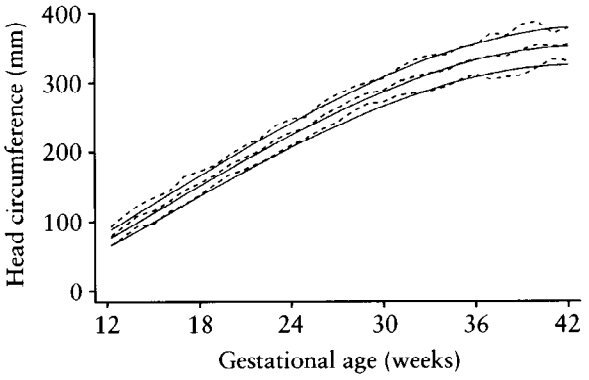


Figure 15 Estimated centile curves from the regression analysis (full lines) and the grouped approach (broken lines) for head circumference data

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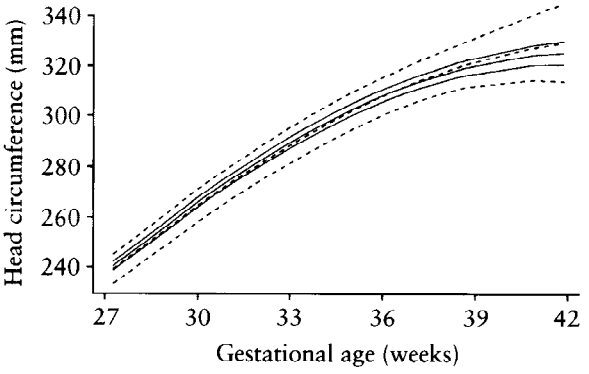


Figure 16 Illustration of inadequate sample size using head circumference data, with 2.5th centile shown for analysis of full dataset (full lines) and reduced dataset (broken lines)

where K is the corresponding centile of the standard Gaussian distribution (as defined in the section, 'Statistical model') and SE(mean) and SE(SD) are obtained from regression software as the standard error of the respective fitted curves (also called 'standard error of the predicted mean'). An approximate 95% confidence band for a given centile may be calculated as plus or minus twice the SE. Note that the SE is itself a curve, that is the precision of the centile curves varies with gestational age. In general, precision is greatest near the middle of the observed age range.

The 95% reference interval for the head circumference data was illustrated in Figure 7. We recall that the sample size for this dataset is 594. A random sample of 50 of these data was selected and the 95% interval and mean curve estimated. Ninety-five per cent confidence bands for the 2.5th centiles for the full and reduced datasets are shown in Figure 16 for gestational age greater than 27 weeks. (Note the reduced width in scale of the axes compared to those of Figure 15.) The difference between the estimated centile curves from the full and reduced data is relatively minor, but the confidence band is much wider for the reduced dataset than for the full one.

Royston[2] gives some suggestions as to choice of a suitable sample size.

DISCUSSION

In our experience, the method described here is adequate for most fetal measurements. The key ideas are the fitting of appropriate curves to the mean and standard deviation. A log transformation of the measurement of interest will often reduce the complexity of curve shape required for the standard deviation, as well as reducing skewness. It is important to assess aspects of the model fit, particularly the distribution of the Z scores. An impression of the precision of the final centile estimates may be obtained from approximate confidence bands[2].

Fractional polynomials were described by Royston and Altman[7] and have been applied to fetal measurements (e.g. references 1, 4 and 6). These regression models have been shown to provide more flexible curve shapes than standard polynomials while also requiring fewer terms. In the examples in the present paper, two-term fractional polynomials appear to have curve shapes similar to cubic polynomials. However, clinically implausible features in conventional high-order (cubic or quartic) polynomials are more likely to occur than with fractional polynomials.

Estimates of the mean and SD curves may be refined using a 'generalized least squares' algorithm. This is an iterative procedure in which the fitted SD curve is used to provide weights for the next cycle of fitting of the mean curve. Although, in practice, there is usually very little improvement in fit, the approach may be useful when the SD curve has a complex shape (E. M. Wright and P. Royston, submitted for publication).

The techniques employed in this paper are least squares estimation (regression) and simple transformations. These are available in even the most basic statistical packages. E. Wright and P. Royston (submitted for publication) describe a program written for the package Stata 5.0[10] which carries out the method described here. Other packages may have alternative methods of estimating age-specific reference intervals (or quantile curves) built into them, although we are unaware of any.

There are some cases that cannot be handled properly by the present approach. Skewed data (i.e. data with an asymmetric distribution) may sometimes be corrected by a log transformation, but this is not invariably successful. E. M. Wright and P. Royston (submitted for publication) describe an extension of the present method to estimate the skewness, using a scaled exponential transformation. Techniques are also available for cases in which the curve shape is complex[11] or the tails of the distribution are non-Gaussian[12]. There are rare occasions in which non-parametric estimation methods are necessary, but we prefer to use such curves to give initial subjective impressions prior to the application of parametric methods.

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APPENDIX: FRACTIONAL POLYNOMIALS

A conventional polynomial is of the form

$$a + b \cdot \text{age} + c \cdot \text{age}^2 + \ldots$$

where a, b, c, ... are the coefficients. A fractional polynomial[7] is a straightforward generalization of this formula:

$$a + b \cdot \text{age}^p + c \cdot \text{age}^q + \ldots$$

where p, q, ... are numbers selected from the set {−2, −1, −0.5, 0, 0.5, 1, 2, 3}. Conventionally, the powers are placed in ascending order and the power 0 represents a natural logarithmic transformation, so that age^0 equals $\log_e(\text{age})$. For example, a fractional polynomial with powers (0,2) (that is, with $p = 0$ and $q = 2$) is of the form

$$a + b \cdot \log_e(\text{age}) + c \cdot \text{age}^2$$

An important extension, which can be justified mathematically, involves 'repeated powers'. In this case the second term is multiplied by $\log_e(\text{age})$. For example, a fractional polynomial with powers (2,2) is of the form

$$a + b \cdot \text{age}^2 + c \cdot \text{age}^2 \cdot \log_e(\text{age})$$

Finding the best model

A fractional polynomial of first degree is of the form $a + b \cdot \text{age}^p$. For a given dataset, the best value of p is found by fitting eight separate linear regressions using age^{-2}, age^{-1}, ..., age^3 and selecting the value of p which gives the best fit (i.e. the lowest residual standard deviation). To find the best-fitting fractional polynomial of second degree, the 36 regression models of the form $a + b \cdot \text{age}^p + c \cdot \text{age}^q$, representing all possible combinations of p and q, are tried. The combination which results in the lowest residual standard deviation is again chosen as the best model.

An example of fractional polynomial analysis is shown in Table 2, which gives the results of fitting all possible first- and second-degree models to the head circumference data. The best-fitting first-degree model has power $p = 0$ and a residual SD of 11.4 mm, whereas the best second-degree model has powers $p = 2$, $q = 2$ and a residual SD of 10.1 mm, a reduction of 11%. Note that several second-degree models have a residual SD not much larger than that of the best model. This is a typical feature of such an analysis and it implies that there is some choice in the final model. Often the one with the most appealing curve shape may be selected. In this case we might consider choosing among the models with powers (−2,−1), (1,2), (1,3), (2,2) and (2,3).

Which degree?

It is reasonable to ask whether the best second-degree model is a significantly better fit than the best first-degree model. To answer this question, Royston and Altman[7] proposed an approach based on the 'deviances' of the different models. A conceptually easier alternative is as follows. If the power p from the best first-degree model appears also in the best second-degree model, one may compare the coefficient of the 'other' term in the second-degree model with its standard error, exactly as when testing a term, such as the cubic coefficient, in a standard polynomial regression model. If the 'other' term is less than

Table 2 Fractional polynomial models for mean head circumference as a function of gestational age. Values in the body of the table are residual standard deviations

First degree		*Second degree*								
		Powers			*Powers*			*Powers*		
Power p	*Residual SD* (mm)	p	q	*Residual SD* (mm)	p	q	*Residual SD* (mm)	p	q	*Residual SD* (mm)
−2	29.6	−2	−2	10.7	−1	1	12.1	0	2	11.4
−1	18.5	−2	−1	10.5	−1	2	12.9	0	3	11.4
−0.5	13.8	−2	−0.5	10.8	−1	3	13.7	0.5	0.5	11.3
0	11.4	−2	0	11.3	−0.5	−0.5	11.2	0.5	1	11.1
0.5	12.8	−2	0.5	11.9	−0.5	0	11.4	0.5	2	10.8
1	16.9	−2	1	12.7	−0.5	0.5	11.5	0.5	3	10.6
2	26.8	−2	2	14.6	−0.5	1	11.7	1	1	10.8
3	35.9	−2	3	16.4	−0.5	2	12.1	1	2	10.4
		−1	−1	10.8	−0.5	3	12.4	1	3	10.2
		−1	−0.5	11.0	0	0	11.4	2	2	10.1
		−1	0	11.3	0	0.5	11.4	2	3	10.4
		−1	0.5	11.7	0	1	11.4	3	3	11.7

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twice its standard error, the second-degree model is rejected in favor of the first-degree model.

When the powers from the best first- and second-degree models are all different (the more common case), a new model with terms corresponding to all three powers is fitted. The two terms from the second-degree model are tested jointly for statistical significance using an F test. This is a standard statistical procedure to be found in most packages which can fit multiple regression models. If the F test is not significant, the second-degree model is rejected in favor of the first-degree model. A similar procedure may be used to compare different models of the same degree, as illustrated below.

For the head circumference data, the best first- and second-degree models have powers (0) and (2,2) respectively. The F test of the terms age^2 and age$^2 \cdot \log_e$(age) in the model which also includes $\log_e$(age) has $F = 82.94$ on (2,590) degrees of freedom, which is highly significant ($p < 0.001$). Thus, the second-degree model fits significantly better than the first-degree model. The next best-fitting model has powers (1,3) (see Table 2). When a model with powers (1,3) and (2,2) is fitted, the F test for the terms age^2 and age$^2 \cdot \log_e$(age) in this model has $F = 6.29$ on (2,589) degrees of freedom, for which $p = 0.002$. The model with powers (2,2) therefore fits significantly better than that with powers (1,3), despite the small difference of only 0.1 mm between the residual SDs. In fact, the curves from the two models are very similar, illustrating the key point that a statistically significant effect may lack practical importance, particularly for large samples.

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